SQL 5-Minute Cheat Sheet for Data Analysts

1. SQL Basics Flow

```
Create Database → Use Database → Create Table → Insert Data → Select Data →
Filter → Sort → Analyze
```

---

2. CREATE DATABASE

What It Does

Creates a new database.

Syntax

```sql
CREATE DATABASE company_db;
```

Key Points

- Database = storage container
- Holds tables
- Used at beginning of projects

Common Mistakes

- Thinking database automatically gets selected
- Forgetting to use:

```sql
USE company_db;
```

---

3. CREATE TABLE

What It Does

Creates table structure.

Syntax

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    name VARCHAR(50),
    salary DECIMAL(10,2)
);
```

Key Points

- Table stores rows and columns
- Data types define data kind
- Constraints control data quality

Common Mistakes

- Wrong datatype selection
- Forgetting primary key
- Using very small VARCHAR size

---

4. DATA TYPES

Data Type	Used For
INT	Whole numbers
DECIMAL	Money/decimal values
VARCHAR	Text
DATE	Dates
BOOLEAN	True/False

Important Difference

CHAR	VARCHAR
Fixed size	Variable size
Wastes extra space	Saves space

Example

```
name VARCHAR(50)
salary DECIMAL(10,2)
```

Common Mistakes

- Using INT for phone numbers
- Storing dates as text

---

5. CONSTRAINTS

Constraint	Purpose
NOT NULL	Value required
UNIQUE	No duplicates
PRIMARY KEY	Unique row ID
FOREIGN KEY	Connect tables
CHECK	Apply condition
DEFAULT	Default value

Example

```
emp_id INT PRIMARY KEY,
age INT CHECK(age >= 18)
```

Common Mistakes

- Duplicate primary keys
- Missing NOT NULL values

---

6. INSERT

What It Does

Adds rows into table.

Syntax

```
INSERT INTO employees (emp_id, name, salary)
VALUES (101, 'Rahul', 50000);
```

Multiple Rows

```
INSERT INTO employees
VALUES
(101, 'Rahul', 50000),
(102, 'Ritu', 60000);
```

Key Points

- Text values need quotes
- Column order matters

Common Mistakes

- Forgetting commas
- Wrong datatype order
- Missing quotes for text

---

7. UPDATE

What It Does

Changes existing data.

Syntax

```
UPDATE employees
SET salary = 60000
WHERE emp_id = 101;
```

Key Points

- WHERE is very important
- Without WHERE all rows update

Common Mistakes

```sql
UPDATE employees
SET salary = 60000;
```

This updates every row.

---

8. DELETE

What It Does

Removes rows from table.

Syntax

```sql
DELETE FROM employees
WHERE emp_id = 101;
```

Key Points

- Removes rows only
- Table still exists

Common Mistakes

```sql
DELETE FROM employees;
```

Deletes all rows.

---

9. ALTER TABLE

What It Does

Changes table structure.

Add Column

```sql
ALTER TABLE employees
ADD COLUMN email VARCHAR(100);
```

Drop Column

```sql
ALTER TABLE employees
DROP COLUMN email;
```

Modify Column

```sql
ALTER TABLE employees
MODIFY COLUMN name VARCHAR(100);
```

Key Points

- Changes structure, not data

Common Mistakes

- Dropping important columns accidentally

---

10. SELECT

What It Does

Retrieves data from table.

Syntax

```sql
SELECT name
FROM employees;
```

Select All

```
SELECT *
FROM employees;
```

DISTINCT

```
SELECT DISTINCT department
FROM employees;
```

Key Points

- Most used command for analysts
- Used for analysis/reporting

Common Mistakes

- Overusing SELECT *

---

11. WHERE CLAUSE

What It Does

Filters rows.

Syntax

```
SELECT *
FROM employees
WHERE salary > 50000;
```

Key Points

- Used heavily in business analysis
- Filters meaningful data

Common Mistakes

- Missing quotes for text values

Wrong:

```
WHERE city = Mumbai
```

Correct:

```
WHERE city = 'Mumbai'
```

---

12. OPERATORS

Arithmetic Operators

Operator	Meaning
+	Add
-	Subtract
*	Multiply
/	Divide

Example

```
SELECT price * quantity AS revenue
FROM sales;
```

Comparison Operators

Operator	Meaning
=	Equal
>	Greater than
<	Less than
>=	Greater than equal
<=	Less than equal
!=	Not equal

Example

```
WHERE salary >= 50000
```

---

13. LOGICAL OPERATORS

AND

```
WHERE city = 'Mumbai'
AND spending > 5000
```

OR

```
WHERE city = 'Mumbai'
OR city = 'Delhi'
```

NOT

```
WHERE NOT department = 'HR'
```

IN

```
WHERE city IN ('Mumbai', 'Delhi')
```

BETWEEN

```
WHERE salary BETWEEN 50000 AND 100000
```

LIKE

```
WHERE name LIKE 'R%'
```

Symbol	Meaning
%	Any characters
_	Single character

Common Mistakes

- Using = NULL instead of IS NULL

---

14. NULL HANDLING

IS NULL

```
WHERE email IS NULL
```

IS NOT NULL

```
WHERE email IS NOT NULL
```

COALESCE

```
COALESCE(email, 'No Email')
```

Replaces NULL values.

NULLIF

```
NULLIF(a,b)
```

Returns NULL if values are equal.

Common Mistakes

Wrong:

```
WHERE email = NULL
```

Correct:

```sql
WHERE email IS NULL
```

---

15. ORDER BY

What It Does

Sorts data.

ASC

```sql
ORDER BY salary ASC
```

Low → High

DESC

```sql
ORDER BY salary DESC
```

High → Low

Common Mistakes

- Forgetting DESC for highest values

---

16. LIMIT

What It Does

Limits number of rows shown.

Syntax

```
SELECT *
FROM employees
LIMIT 5;
```

Real Use

- Testing queries
- Viewing sample data
- Dashboard previews

Common Mistakes

Thinking LIMIT deletes rows. It only limits output.

---

17. Most Important Real-World Analyst Concepts

Analysts Mostly Use:

- SELECT
- WHERE
- ORDER BY
- LIMIT
- GROUP BY (next topic)
- JOINS (very important later)

Real Analyst Work Includes:

- filtering business data
- finding trends
- analyzing revenue
- customer segmentation
- dashboard creation
- KPI reporting
- data cleaning

Final Quick Revision

Command	Purpose
CREATE DATABASE	Create database
CREATE TABLE	Create table
INSERT	Add rows
UPDATE	Modify rows
DELETE	Remove rows
ALTER TABLE	Change structure
SELECT	Retrieve data
WHERE	Filter data
ORDER BY	Sort data
LIMIT	Restrict output
DISTINCT	Remove duplicates
COALESCE	Replace NULL
LIKE	Pattern search
BETWEEN	Range filtering
IN	Multiple matching

Golden Rule for SQL Analysts

Do not memorize syntax blindly.

Always think:

1. What business problem am I solving?
2. What data do I need?
3. How should I filter it?
4. What insight will help decision-making?

That mindset separates real analysts from students who only memorize queries.